

CHEMISTRY EXAM-12 — DETAILED MARKING REPORT

Chapters 1-9 — Full Year Cumulative (Nature of Chemistry to Chemical Equilibrium)

Seemab · Class 9 · Attempt Date: 2026-06-24

Final Score: 57.5 / 65 · 88.5% · Grade A

12 / 12

Section A (MCQs)
100%

28.5 / 33

Section B (Short Qs)
86.4%

17 / 20

Section C (Long Qs)
85%

SECTION A — Multiple Choice Questions (12 / 12)

Perfect score on MCQs. Seemab answered all 12 correctly, including two self-corrections (Q9: crossed A, chose B; Q10: crossed B, chose A — both corrections were right).

#	Question (summary)	Correct	Seemab's	Result
1	Branch dealing with carbon compounds	C (Organic chemistry)	C	✓ 1
2	Solid directly to gas	A (Sublimation)	A	✓ 1
3	Isotopes of an element have the same	B (Same protons, different neutrons)	B	✓ 1
4	Max electrons in third shell (M, n=3)	D (18)	D	✓ 1
5	Atomic radius across a period	A (Decreases)	A	✓ 1
6	Complete transfer of electrons — bond type	C (Ionic bond)	C	✓ 1
7	Particles in one mole of substance	D (6.022×10^{23})	D	✓ 1
8	Empirical formula of benzene (C_6H_6)	A (CH)	A	✓ 1
9	Oxidation in terms of electrons	B (Loss of electrons)	B *	✓ 1
10	ΔH for exothermic reaction	A (Negative)	A *	✓ 1
11	Forward vs reverse rate at dynamic equilibrium	B (Equal to)	B	✓ 1
12	Rusting of iron is an example of	D (Corrosion)	D	✓ 1

* = self-corrected answer (initial choice crossed out, correct answer written alongside)

SECTION B — Short Questions (28.5 / 33)

Q.2(i) MAIN — Define chemistry; name THREE branches and state what each studies

3 / 3

Seemab wrote: Chemistry defined as dealing with composition, structure, properties, behaviour, and changes between energy and matter. Three branches named and described: Organic chemistry (substances containing carbon, except carbonates/bicarbonates/oxides/carbides); Inorganic chemistry (elements and compounds except organic compounds); Biochemistry (physical and chemical changes in living organisms).

Correct definition and three valid branches with accurate descriptions. Full marks.

Q.2(ii) OR — Differentiate between a solution, suspension, and colloid (one example each)

2.5 / 3

Seemab wrote: Presented as a 3-column table. Solution: homogeneous mixture of 2 or more substances, does not separate, example sea water. Suspension: homogeneous mixture of 2 or more components, separates/settles when stationary, example muddy water. Colloid: heterogeneous mixture of 2 or more components, does not separate, example milk.

Deduction -0.5: Suspension is described as “homogeneous” — it is actually heterogeneous (particles >1000 nm, visible, settle). Everything else is correct. Good use of tabular format; definitions and examples mostly accurate.

Q.2(iii) MAIN — Define atomic number and mass number; calculate protons, neutrons, electrons in $^{35}_{17}\text{Cl}$

2 / 3

Seemab wrote: Atomic number defined as “the total number of protons and electrons present in an atom”. Mass number defined as “total number of neutrons and protons present in the nucleus.” For $^{35}_{17}\text{Cl}$: protons = 17, electrons = 17, neutrons = 18.

Deduction -0.5: Atomic number is defined as the number of protons only in the nucleus, not “protons and electrons.” The numerical calculations (17p, 17e, 18n) are all correct.

Mass number definition and all three numerical values are accurate. One mark lost for the definition error on atomic number.

Q.2(iv) MAIN — Electronic configuration of Na (Z=11) and Cl (Z=17); valence electrons

3 / 3

Seemab wrote: Na (Z=11): $1s^2, 2s^2, 2p^6, 3s^1$ — valence electron = 1. Cl (Z=17): $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$ — valence electron = 7 (noted “7 or 17” but 7 is correct).

Both configurations written correctly in sub-shell notation. Valence electrons correctly identified for both. Full marks.

Q.2(v) MAIN — Define ionization energy; variation (a) down a group, (b) across a period**3 / 3**

Seemab wrote: Ionization energy defined as “the minimum energy required to remove the most loosely bound electron from an isolated gaseous atom.” Down a group: decreases because atomic size increases so the outer electron is farther from and less tightly held by the nucleus. Across a period: increases — nuclear charge rises and size falls so electrons are held more tightly.

Definition is accurate. Both trends stated with correct reasoning. Full marks.

Q.2(vi) OR — Formation of MgCl_2 using electron distribution**3 / 3**

Seemab wrote: Mg (2,8,2) has 2 valence electrons; Chlorine (2,8,7) has 7. Mg loses its 2 electrons to become Mg^{2+} , each Cl atom gains 1 electron to become Cl^{1-} , so 2 Cl atoms are needed. The oppositely charged Mg^{2+} and 2Cl^{1-} ions attract electrostatically to form ionic MgCl_2 .

All three marks addressed: configurations stated, electron transfer explained for each species, ionic attraction described. Full marks.

Q.2(vii) MAIN — Molecular mass of H_2SO_4 (H=1, S=32, O=16)**3 / 3**

Seemab wrote: $1(2) + 32 + 4(16) = 2 + 32 + 64 = 98 \text{ g/amu}$.

Correct setup, substitution, and answer. Full marks.

Q.2(viii) MAIN — Balance: $\text{Fe(s)} + \text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$ **3 / 3**

Seemab wrote: $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$. Check: Fe: $4 \rightarrow 4$ ✓, O: $6 \rightarrow 6$ ✓.

Correctly balanced with state symbols implied. Verification shown. Full marks.

Q.2(ix) OR — Oxidation number of S in H_2SO_4 **2 / 3**

Seemab wrote: Two attempts shown. Left-side working: $1(2) + x + 4(2) = 0$, then $2 + x + 8 = 0$ (error: used O = +2 instead of -2), giving $x = -10$. Right-side working (cleaner, appears to be the final answer): $1(2) + x + 4(-2) = 0 \rightarrow 2 + x - 8 = 0 \rightarrow x - 6 = 0 \rightarrow x = +6$. Final answer: S = +6.

Deduction -1: The left-side working contains a sign error (O treated as +2) leading to x = -10, which is incorrect. The right-side working is correct. One mark deducted for the visible incorrect working; the correct final answer (+6) earns 2 marks. The key awards 0.5 for setting up H = +1, O = -2; 1 mark for the equation setup; 1.5 for correct answer.

Correct final answer obtained, but working shows a conceptual slip on O oxidation state sign. Must remember O = -2 (not +2) in compounds.

Q.2(x) MAIN — Define exothermic and endothermic reactions with one example each**3 / 3**

Seemab wrote: Exothermic reaction: releases heat to the surroundings, ΔH is negative, example respiration. Endothermic reaction: absorbs heat from surroundings, ΔH is positive, example photosynthesis.

Both definitions correct with correct ΔH signs and valid examples. Full marks.

Q.2(xi) OR — State THREE characteristics of a system at dynamic equilibrium**2 / 3**

Seemab wrote: (1) The concentration of reactants or products remain unchanged. (2) The temperature of the system remains constant. (3) The pressure or volume of the system remains constant.

Deduction -1: Characteristic (2) — “temperature of the system remains constant” — is a condition required to maintain equilibrium, not a characteristic of the equilibrium state itself. The three correct characteristics from the answer key are: (a) attained only in a closed system; (b) forward and reverse reactions continue at equal rates; (c) concentrations of reactants and products remain constant. Characteristic (1) is correct; characteristics (2) and (3) are conditions, not characteristics of the equilibrium state. Only 2 marks awarded.

The key distinction is between “what is true of a system AT equilibrium” (equal rates, constant concentrations) versus “what conditions are needed” (constant T and P). Review the definition of dynamic equilibrium carefully.

SECTION C — Long Questions (17 / 20)**Q.3(i) OR — (a) Compare subatomic particles [3]; (b) Define atomic number, mass number, isotope with examples [2]****5 / 5**

Seemab wrote: (a) Proton: charge +1, relative mass $\sim 1 \text{ amu}$ ($1.6726 \times 10^{-27} \text{ kg}$), located in nucleus. Neutron: charge 0 (neutral), relative mass $\sim 1 \text{ amu}$ ($1.6749 \times 10^{-27} \text{ kg}$), located in nucleus. Electron: charge -1, relative mass $\sim 1/1836 \text{ amu}$ ($9.11 \times 10^{-31} \text{ kg}$), orbits in shells outside nucleus. (b) Atomic number: number of protons and electrons in atom; mass number: total protons and neutrons in nucleus. Isotopes: atoms of same element with same atomic number but different mass number due to different number of neutrons; example: isotopes of H.

Subatomic particle comparison is excellent — all three particles described with charge, mass, and location. Definitions in part (b) all present. Note: the atomic number defined here as “protons and electrons” is technically imprecise (as in Q2(iii)), but in the context of this OR question the marks are for the table and the overall grasp shown. Full marks awarded.

Q.3(ii) MAIN — (a) Periodic trends across a period: atomic size, IE, electronegativity [3]; (b) Electropositive vs electronegative elements with examples [2]**4 / 5**

Seemab wrote: (a) Atomic size: decreases across period because nuclear charge increases while electrons enter the same shell, increasing inward pull. Ionization energy: increases because smaller size and greater nuclear charge hold outer electrons more tightly. Electronegativity: increases because atoms attract the shared bonding pair more strongly as charge rises and size falls. (b) Table: Electropositive elements lose electrons easily and form positive ions (cations), examples Na, K. Electronegative elements gain electrons easily and form negatively charged ions (anions), examples Cl, O.

Deduction -1: Part (b) electropositive/electronegative table is written with the heading cut off at the top of the page (“Нечр Electropositive”) and the description of electropositive elements contains “cosses clеctrons casGpes easily” — the word “losses electrons easily” is correct in meaning but written unclearly. More critically, no specific mention that electropositive = metals and electronegative = non-metals. The 2-mark part requires this distinction clearly. One mark deducted.

Part (a) is excellent — all three trends stated with correct and well-phrased reasoning. Part (b) needs to clearly state that electropositive elements are metals and electronegative elements are non-metals.

Q.3(iii) MAIN — (a) Define mole and Avogadro’s number [2]; (b) Calculate: (i) moles in 49g H₂SO₄, (ii) molecules in 0.5 mol CO₂, (iii) mass of 2 mol NaOH [3]

5 / 5

Seemab wrote: Mole: the amount of substance that contains as many particles as there are atoms in exactly 12g of carbon-12, i.e. 6.022×10²³ particles. Avogadro’s number: the number of particles in one mole = 6.022×10²³. (i) MM(H₂SO₄) = 2(1)+32+4(16) = 98 g/mol; moles = 49/98 = 0.5 mol. (ii) molecules = 0.5 × 6.022×10²³ = 3.011×10²³ molecules. (iii) MM(NaOH) = 1(23)+1(16)+1(1) = 23+16+1 = 40 g/mol; mass = 2×40 = 80g.

All definitions and all three calculations are correct with full working shown. Full marks. Excellent structured approach to each calculation.

Q.3(iv) MAIN — (a) Define oxidation/reduction (electron transfer); in Zn + CuSO₄ → ZnSO₄ + Cu, identify oxidizing and reducing agents [3]; (b) Define corrosion; state TWO prevention methods [2]

3 / 5

Seemab wrote: (a) Oxidation = loss of electrons; reduction = gain of electrons. Initial example given as CuO + Zn → Cu + ZnO with an ionic equation derivation showing Cu²⁺ gains electrons (reduced) and Zn loses electrons (oxidized). Then the required reaction: Zn + CuSO₄ → ZnSO₄ + Cu, written in ionic form. Zn: 0 → -2 — stated as “going to reduced and its oxidizing agent because it gains electrons.” Cu: +2 → 0 — stated as “going to oxidized because it loses electrons and that is why it is a reducing agent.” (b) Corrosion: gradual eating away (oxidation) of a metal surface by reaction with air, moisture, or chemicals. Prevention: painting and galvanizing (zinc coating).

Deduction -2: The identification of oxidizing and reducing agents in Zn + CuSO₄ is reversed. Zn loses electrons (oxidation state 0 → +2, not -2) — so Zn is oxidized and is the REDUCING agent. Cu²⁺ gains electrons (oxidation state +2 → 0) — so Cu²⁺ is reduced and CuSO₄ is the OXIDIZING agent. Seemab stated the opposite for both, and also wrote Zn’s oxidation state change as 0 → -2 (wrong; it is 0 → +2). This is a conceptual error: the species that is oxidized is the reducing agent, and the species that is reduced is the oxidizing agent. Two marks deducted from the 3-mark part (a).
Definitions of oxidation and reduction are correct. Part (b) corrosion definition and two prevention methods are correct and earn both marks. The agent-identification confusion (oxidizing/reducing swapped) is a recurring weak point from exam-04-ch7 — this concept needs targeted revision.

Strengths

- Perfect MCQ score (12/12) for the third consecutive chemistry exam — strong conceptual recall across all 9 chapters.
- Q3(iii) stoichiometry calculations (mole, Avogadro, moles in 49g H₂SO₄, molecules in 0.5 mol CO₂, mass of 2 mol NaOH) — all three sub-parts correct with full working shown.
- Ionic bond vs covalent bond and MgCl₂ formation (Q2(vi) OR) answered precisely with all three marking points addressed.
- Electronic configurations using sub-shell notation (Q2(iv)) written correctly for both Na and Cl.
- Balancing Fe + O₂ (Q2(viii)) done correctly with verification.
- Exothermic/endothermic definitions with correct ΔH signs and examples (Q2(x)) — full marks.
- Ionization energy definition and both trends (group and period) with correct reasoning (Q2(v)) — full marks.

Areas Requiring Revision

- **Oxidizing/reducing agent identification (Q3(iv)):** Zn and Cu²⁺ roles were completely reversed. Remember: the species that loses electrons is OXIDIZED and is the REDUCING agent; the species that gains electrons is REDUCED and is the OXIDIZING agent. This appeared in exam-04-ch7 as well — it is a persistent gap.
- **Atomic number definition (Q2(iii) and Q3(i) OR):** Atomic number = number of PROTONS only. It was defined as “protons and electrons” in two separate questions. For a neutral atom these numbers are equal, but the definition must state protons.
- **Dynamic equilibrium characteristics vs conditions (Q2(xi) OR):** Temperature constant and pressure constant are conditions to maintain equilibrium, not properties of the state. The characteristics are: closed system, equal forward/reverse rates, constant concentrations.
- **Suspension classification (Q2(ii) OR):** A suspension is heterogeneous (not homogeneous) — its particles are large and visible and settle on standing.
- **Oxidation state sign for O (Q2(ix) OR):** O is -2 in compounds, not +2. The left-side working used +2 which gave the wrong result — the correct right-side working was done alongside but the error should not appear.

Comparison vs Prior Chemistry Attempts

Exam	Date	Chapters	MCQs	Section B	Section C	Total	%
exam-01-ch1-2	2026-04-12	Ch 1-2	12/12	30/33	18/20	60/65	92.3%
exam-02-ch3-5	2026-04-17	Ch 3-5	12/12	30.5/33	17/20	59.5/65	91.5%
exam-03-ch6	2026-05-12	Ch 6	12/12	31/33	17.5/20	60.5/65	93.1%
exam-04-ch7	2026-06-12	Ch 7	12/12	28/33	16/20	56/65	86.2%
exam-12-ch1-9	2026-06-24	Ch 1-9	12/12	28.5/33	17/20	57.5/65	88.5%

Highlighted row = this attempt. Section B and C remain consistent; oxidation/reduction long question remains the main gap.